



Dresden Nuclear Power Station
6500 North Dresden Road
Morris, IL 60450

SVPLTR # 19-0067

10 CFR 50.73(a)(2)(i)(B)

November 19, 2019

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Dresden Nuclear Power Station, Unit 3
Renewed Facility Operating License No. DPR-25
NRC Docket No. 50-249

Subject: Licensee Event Report 249/2019-001-00, Degraded Differential Pressure
Indicating Switch in the LPCI Loop Select Logic Caused by Water Intrusion
Results in a Condition Prohibited by Technical Specifications

Enclosed is Licensee Event Report 249/2019-001-00, "Degraded Differential Pressure
Indicating Switch in the LPCI Loop Select Logic Caused by Water Intrusion Results in a
Condition Prohibited by Technical Specifications." This report describes an event which is being
reported in accordance with 10 CFR 50.73(a)(2)(i)(B), "Any operation or condition which was
prohibited by the plant's Technical Specifications."

There are no regulatory commitments contained in this submittal.

Should you have any questions concerning this letter, please contact Mr. Bruce Franzen
at (815) 416-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "Peter J. Karaba".

Peter J Karaba
Site Vice President
Dresden Nuclear Power Station

Enclosure Licensee Event Report 249/2019-001-00

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Dresden Nuclear Power Station

IEZZ
NRR



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R 3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-mv/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NE08-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Dresden Nuclear Power Station, Unit 3

2. Docket Number

05000249

3. Page

1 OF 4

4. Title

Degraded Differential Pressure Indicating Switch in the LPCI Loop Select Logic Caused by Water Damage Results in a Condition Prohibited by Technical Specifications

5. Event Date

Month	Day	Year
09	20	2019

6. LER Number

Year	Sequential Number	Rev No
2019	001	00

7. Report Date

Month	Day	Year
11	19	2019

8. Other Facilities Involved

Facility Name	Docket Number
N/A	N/A

9. Operating Mode

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

1

☐ 20 2201(b)	☐ 20 2203(a)(3)(i)	☐ 50 73(a)(2)(ii)(A)	☐ 50 73(a)(2)(vii)(A)
☐ 20 2201(d)	☐ 20 2203(a)(3)(ii)	☐ 50 73(a)(2)(ii)(B)	☐ 50 73(a)(2)(vii)(B)
☐ 20 2203(a)(1)	☐ 20 2203(a)(4)	☐ 50 73(a)(2)(iii)	☐ 50 73(a)(2)(ix)(A)
☐ 20 2203(a)(2)(i)	☐ 50 36(c)(1)(i)(A)	☐ 50 73(a)(2)(iv)(A)	☐ 50 73(a)(2)(x)
☐ 20 2203(a)(2)(ii)	☐ 50 36(c)(1)(ii)(A)	☐ 50 73(a)(2)(v)(A)	☐ 73 71(a)(4)
☐ 20 2203(a)(2)(iii)	☐ 50 36(c)(2)	☐ 50 73(a)(2)(v)(B)	☐ 73 71(a)(5)
☐ 20 2203(a)(2)(iv)	☐ 50 48(a)(3)(ii)	☐ 50 73(a)(2)(v)(C)	☐ 73 77(a)(1)
☐ 20 2203(a)(2)(v)	☐ 50 73(a)(2)(i)(A)	☐ 50 73(a)(2)(v)(D)	☐ 73 77(a)(2)(i)
☐ 20 2203(a)(2)(vi)	☒ 50 73(a)(2)(i)(B)	☐ 50 73(a)(2)(vii)	☐ 73 77(a)(2)(ii)
	☐ 50 73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

10. Power Level

100

12. Licensee Contact for this LER

Licensee Contact

Bruce Franzen

Telephone Number (Include Area Code)

815-416-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	AD	PDIS	I204	Y					

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On September 20, 2019, one of four channels for the 3A reactor recirculation pump high differential pressure function failed the channel functional test. Technical Specification 3.3.5.1, "ECCS Instrumentation," requires all four channels to be operable. Technical Specification 3.3.5.1, Condition C identifies a Completion Time of 24 hours to restore the failed channel to operable status. The degraded differential pressure switch was replaced. The channel was restored to operable within the 24 hour time limit.

On June 20, 2019, the Unit 3 East Low Pressure Coolant Injection Room cooler was found leaking water onto the panel where recirculation pump differential pressure switches are installed. On July 30, 2019, a surveillance documented that all four channels were functional. The failed channel was determined to be inoperable between July 30 and September 20, 2019, which is longer than the 24 hours allowed by Technical Specification. This event is being reported pursuant to 10 CFR 50.73(a)(2)(i)(B), a condition which was prohibited by the plant's Technical Specifications. The cause of the degraded differential pressure switch was due to water dripping from the Low Pressure Coolant Injection room cooler onto the instrumentation rack. Service Water is isolated to the Unit 3 East Low Pressure Coolant Injection Room cooler until the room cooler is repaired.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollcta.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Dresden Nuclear Power Station, Unit 3	05000249	2019	- 001	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

Dresden Nuclear Power Station (DNPS), Unit 3, is a General Electric Company Boiling Water Reactor with a licensed maximum power level of 2957 megawatts thermal. The Energy Industry Identification System codes used in the text are identified as [XX].

A. Plant Conditions Prior to Event:

Unit: 03	Event Date: 09/20/19	Event Time: 1215 CDT
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 100 percent

B. Description of Event:

On September 20, 2019, at 1215 Central Time, one of four channels [CHA] for the 3A recirculation [AD] pump [P] high differential pressure function failed the channel functional test during the performance of Dresden Surveillance procedure DIS 1500-09, "LPCI Reactor Recirculating Pump A and B Differential Pressure Indicating Switch Channel Calibration and Channel Functional Test."

Technical Specification (TS) 3.3.5.1, "ECCS Instrumentation," Table 3.3.3.5.1-1, "Emergency Core Cooling System Instrumentation," identifies four channels per recirculation pump are required for the Low Pressure Coolant Injection System (LPCI) [BO] recirculation pump high differential pressure function during Modes 1, 2, and 3. The recirculation pump high differential pressure function supports the LPCI Loop Select Logic to identify and direct LPCI flow to the unbroken recirculation loop if a Design Basis Accident (DBA) occurs.

TS Limiting Condition for Operation (LCO) 3.3.5.1, "The ECCS instrumentation for each Function in Table 3.3.5.1-1 shall be OPERABLE," was not met. TS 3.3.5.1, Condition C was entered for the failed channel of the recirculation pump high differential pressure function. A TS Completion Time of 24 hours is specified for TS 3.3.5.1, Required Action C, which requires the channel to be restored to an operable status.

During the investigation of the event on September 20, 2019, a differential pressure indicating switch (DPIS) [PDIS] was found degraded. On September 21, 2019, at approximately 1000 Central Time, the DPIS was replaced and TS LCO 3.3.5.1 was declared met. The channel to the recirculation pump high differential pressure function was restored to operable within the 24-hour TS Completion Time and the LPCI Loop Select Logic was restored.

Prior to the event, on June 20, 2019, the Unit 3 East LPCI Room cooler [CLR] identified as leaking water onto the electrical panel where recirculation pump differential pressure switches are installed. A deflector was installed to prevent water from dripping onto the electrical panel. After this leak was discovered, an electrical ground was identified on the Unit 3 125 VDC battery system [EJ]. The DPIS that was found degraded during the September 20, 2019, event was not qualified for spray or submergence.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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05000249

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YEAR	SEQUENTIAL NUMBER	REV NO.
2019	- 001	- 00

NARRATIVE

Prior to the event, the channel functional test surveillance on July 30, 2019, determined that the LPCI recirculation pump high differential pressure was functional with all required channels operable.

On September 20, 2019, at 2145, the degraded DPIS was removed from service. When the wires were removed from the DPIS, the 125 VDC battery electrical ground issue was resolved. It is concluded that the degraded DPIS was not operable after July 30, 2019.

Since the DPIS was not operable after the July 30, 2019, surveillance, one channel of the recirculation pump high differential pressure function was not operable between July 30 and September 20, 2019, (i.e., approximately 52 days). This duration is longer than the TS 3.3.5.1, Condition C, Completion Time of 24 hours. Therefore, this event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B), any condition which was prohibited by the plant's Technical Specifications.

C. Cause of Event:

The cause of the degraded DPIS was due to water dripping from the Unit 3 East LPCI Room cooler onto the instrumentation rack where the DPIS was installed.

D. Safety Analysis:

Recirculation Pump Differential Pressure signals are used by the LPCI Loop Select Logic to determine if either recirculation pump is running. Recirculation Pump Differential Pressure - High (i.e., Break Detection) signals are initiated from eight differential pressure switches, four of which sense the pressure differential between the suction and discharge of each reactor recirculation pump.

The LPCI System initiation logic contains LPCI Loop Select Logic to identify and direct LPCI flow to the unbroken recirculation loop if a DBA occurs. The LPCI Loop Select Logic is initiated upon the receipt of an Emergency Core Cooling System [JE] initiation signal. When initiated, the LPCI Loop Select Logic first determines recirculation pump operation by sensing the differential pressure between the suction and discharge of each pump.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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NARRATIVE

There are four DPISs monitoring each recirculation pump which are, in turn, connected to relays whose contacts are connected to two trip systems. The contacts are arranged in a one-out-of-two taken twice logic for each recirculation pump. There are two redundant trip systems in the LPCI Loop Select Logic. The complete logic in each trip system must actuate for operation of the LPCI Loop Select Logic.

Although the single DPIS was not functional as discovered on September 20, 2019, there was no impact to the LPCI system. The failure of the DPIS did not impact the ability of the LPCI system to perform its safety function in the event of an accident. Failure of one switch impacts one division. However, the unaffected division remains capable of performing function. There was no loss of safety function.

The other seven LPCI Loop Select Logic DPISs remained functional. The three other DPISs installed in the same electrical panel remained operable since there were no additional 125 VDC system electrical grounds. The four other DPISs related to LPCI Loop Select Logic are installed in a separate panel and no water was found to be dripping on this panel.

E. Corrective Actions:

TS LCO 3.3.5.1, Condition C was entered for the event on September 20, 2019. Following DPIS replacement and testing, the channel of recirculation pump high differential pressure function was restored to operable within the 24-hour Completion Time of TS 3.3.5.1, Condition C.

Service Water is currently isolated to the Unit 3 East Low Pressure Coolant Injection Room cooler, and will remain so until the room cooler is repaired. The LPCI Room cooler is not required for the operability of the LPCI system.

F. Previous Occurrences:

No previous occurrences were identified associated with this type of event.

G. Component Failure Data:

Barton Differential Pressure Indicator Switch, Model 288.